

DNA Sequence Analysis Tool Using Advanced String-Matching Algorithms

DSA-III Mini Project

Abstract

The DNA Sequence Analysis Tool is a Java-based web application developed as a DSA-III mini-project for efficient DNA motif searching and performance analysis. The system implements four classical string-matching algorithms: Knuth-Morris-Pratt (KMP), Z-Algorithm, Rabin-Karp, and Aho-Corasick. KMP, Z-Algorithm, and Rabin-Karp search for individual motifs, while Aho-Corasick enables simultaneous searching of multiple motifs.

The application processes DNA sequences composed of four nucleotide bases: A – Adenine, T – Thymine, G – Guanine, and C – Cytosine. Users can enter a DNA sequence manually or upload .txt/.fasta files and search for specific DNA motifs. The system identifies motif occurrences, displays their exact positions, highlights matching regions, measures the execution time of each algorithm, and identifies the fastest-performing algorithm for a given search. It also compares the algorithms based on their time and space complexity.

The application operates entirely in memory without requiring a database, providing a lightweight platform for demonstrating the practical application of advanced string-matching algorithms in bioinformatics, genome analysis, and DNA sequence processing.

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